

System and Network Security

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Definition: A replay attack occurs when an adversary captures valid communication messages and retransmits them later to gain unauthorized access.

Key Characteristics:

- No need to know secret keys
- Exploits lack of freshness (timestamps, nonces)
- Common in authentication protocols

Example Scenario:

- User sends login message: ID || Hash(PW)
- Attacker captures the message
- Attacker replays it later to impersonate the user

Impact:

- Unauthorized system access
- Breach of confidentiality

Definition: In a MITM attack, an adversary secretly intercepts and alters communication between two legitimate parties.

Key Characteristics:

- Active attack
- Breaks authentication and key agreement
- Often exploits weak key exchange

Example Scenario:

- User sends public key to server
- Attacker replaces it with their own key
- Attacker decrypts and re-encrypts all messages

Impact:

- Data manipulation
- Loss of confidentiality and integrity

Definition: An impersonation attack occurs when an attacker pretends to be a legitimate entity to gain access.

Key Characteristics:

- Targets authentication mechanisms
- Can involve stolen credentials or protocol flaws

Example Scenario:

- Attacker guesses or steals a user's identity
- Sends authentication requests as the victim

Impact:

- Unauthorized access
- Identity misuse

Definition: A privileged-insider attack is carried out by a trusted entity with authorized system access.

Key Characteristics:

- Hard to detect
- Exploits excessive privileges

Example Scenario:

- System administrator accesses stored passwords
- Misuses credentials to impersonate users

Impact:

- Data leakage
- Trust violation

Denial-of-Service (DoS) Attack

Definition: A DoS attack aims to make a system unavailable to legitimate users by exhausting resources.

Key Characteristics:

- Targets availability
- Can be flooding or logic-based

Example Scenario:

- Attacker sends massive fake requests
- Server becomes overloaded and crashes

Impact:

- Service disruption
- Financial and operational losses

Definition: A stolen verifier attack occurs when an attacker obtains stored password verifiers from a server.

Key Characteristics:

- Exploits poor credential storage
- Common in password-based systems

Example Scenario:

- Server stores `Hash(PW)`
- Attacker compromises the server
- Uses hash to authenticate elsewhere

Impact:

- Credential compromise
- Large-scale impersonation

Types of Attacks on Encrypted Messages

- Ciphertext Only Attack (COA)
- Known Plaintext Attack (KPA)
- Chosen Plaintext Attack (CPA)
- Chosen Ciphertext Attack (CCA)
- Chosen Text Attack (CTA)

- Security attacks exploit protocol and system weaknesses
- Robust authentication and cryptographic techniques are essential
- Defense mechanisms include nonces, timestamps, mutual authentication, and access control